7 The curve C has equation $y = \dfrac{x+2}{x^2+3x+1}$.

(a) Find the equations of the asymptotes of C. [2]

(b) Show that C has no stationary points. [3]

(c) Sketch C, stating the coordinates of the intersections with the axes. [3]

(d) Sketch the curve with equation $y = \left|\dfrac{x+2}{x^2+3x+1}\right|$. [2]

(e) Find in exact form the set of values of x for which $\left|\dfrac{x+2}{x^2+3x+1}\right| > 2$. [4]

Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.